9 Lecture 9 (February 3, 2025, est.): Affine groups and tensor automorphisms

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Our first reconstruction problem is to recover an affine group scheme G from its finite-dimensional representation category together with the forgetful functor. Throughout this lecture k is a field; no finite-type assumption on G is imposed.

Approximation by algebraic groups

9.1 Proposition. *Every finite subset of a commutative Hopf algebra A over k is contained in a Hopf subalgebra finitely generated as a k-algebra.*

Proof. By local finiteness of the regular comodule, the given subset lies in a finite-dimensional subcomodule $V \subseteq A$. Write $\Delta(v_j) = \sum_i v_i \otimes a_{ij}$ in a basis of V. Coassociativity gives

$$\Delta(a_{ij}) = \sum_s a_{is} \otimes a_{sj}.$$

Thus the finite-dimensional span C of the v_j, the a_{ij}, and 1 satisfies $\Delta(C) \subseteq C \otimes C$. Let A_0 be the algebra generated by $C + S(C)$. Since the antipode reverses coproducts, both $\Delta(C)$ and $\Delta(S(C))$ lie in $A_0 \otimes A_0$. For a commutative Hopf algebra $S^2 = \mathrm{id}$, so $S(A_0) \subseteq A_0$. The counit restricts, and A_0 is the desired Hopf subalgebra. ∎

The commutativity assumption matters for this short argument: it is not enough, for a completely arbitrary Hopf algebra, merely to add one image under the antipode.

9.2 Corollary. *Every affine group scheme is an inverse limit of affine algebraic group schemes*

$$G \simeq \varprojlim_i G_i$$

with faithfully flat transition morphisms.

Proof. Write $A = \mathcal{O}(G) = \bigcup_i A_i$, directed by inclusion over its finitely generated Hopf subalgebras. Then $G_i = \operatorname{Spec} A_i$ is algebraic, and

$$\mathrm{Hom}_{k\text{-alg}}(A, R) \simeq \varprojlim_i \mathrm{Hom}_{k\text{-alg}}(A_i, R)$$

for every commutative k-algebra R. An inclusion of commutative Hopf algebras over a field is faithfully flat; applying this to $A_i \subseteq A_j$ gives the final assertion. This faithful-flatness theorem is proved in [Mil12, Chapter VI, Section 11]. ∎

The algebras A_i are finitely generated, not necessarily finite dimensional. Also, faithful flatness of $G_j \to G_i$ does not assert surjectivity of $G_j(R) \to G_i(R)$ for every test algebra R.

The tensor automorphism functor

Let $\omega : \mathrm{Rep}_k(G) \to \mathrm{Vec}_k$ forget the action. For a commutative k-algebra R, put $\omega_R(V) = \omega(V) \otimes_k R$.

9.3 Definition. The group $\underline{\mathrm{Aut}}^{\otimes}(\omega)(R)$ consists of families of invertible R-linear maps $\lambda_V : \omega_R(V) \to \omega_R(V)$ satisfying

1. $(f \otimes \mathrm{id}_R)\lambda_V = \lambda_W(f \otimes \mathrm{id}_R)$ for every morphism $f : V \to W$;
2. $\lambda_{V\otimes W} = \lambda_V \otimes_R \lambda_W$ under the canonical tensor identification;
3. $\lambda_k = \mathrm{id}_R$ on the trivial representation.

Composition is componentwise; scalar extension makes this a group-valued functor of R.

An element $g \in G(R)$ acts on every V_R, and equivariance of morphisms and the diagonal tensor action give a natural group homomorphism

$$G \longrightarrow \underline{\mathrm{Aut}}^{\otimes}(\omega).$$

We shall prove that it is an isomorphism in Lecture 10.

Reduction to a tensor generator

For $X \in \mathrm{Rep}_k(G)$, let $\mathcal{C}_X$ be the full subcategory of subquotients of finite direct sums of tensor products of X and X^*, including the unit. A tensor automorphism of $\omega|_{\mathcal{C}_X}$ is determined by its component on X. Indeed, the component on X^* is determined by evaluation, those on tensor products and sums follow from tensor compatibility and naturality, and those on subquotients follow from injectivity and surjectivity after tensoring with R. Hence

$$\underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X})(R) \hookrightarrow \mathrm{GL}(X \otimes R).$$

Let G_X be the scheme-theoretic image of $G \to \mathrm{GL}(X)$. It is a closed algebraic subgroup. Its coordinate Hopf algebra is generated by the matrix coefficients of X and its dual. A faithful finite-dimensional representation generates the representation category of an algebraic group under tensor products, sums, duals and subquotients; see [Mil12, Chapter VIII, Section 11]. Thus $\mathcal{C}_X$ identifies with $\mathrm{Rep}(G_X)$, and its action gives inclusions

$$G_X(R) \subseteq \underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X})(R) \subseteq \mathrm{GL}(X \otimes R).$$

Naturality gives preservation of every subrepresentation and compatibility with every quotient. For example, if $Y \subseteq V$, the inclusion gives

$$\lambda_V(Y \otimes R) = Y \otimes R.$$

If $t \in V^G$, the morphism $k \to V$, $1 \mapsto t$, gives $\lambda_V(t \otimes 1) = t \otimes 1$.

Finally, as X varies and direct sums provide upper bounds, the Hopf algebras $\mathcal{O}(G_X)$ exhaust $\mathcal{O}(G)$. Indeed, every element belongs to a finite-dimensional regular

subcomodule and is a linear combination of its matrix coefficients by the counit identity. Thus

$$G = \varprojlim_X G_X, \qquad \underline{\mathrm{Aut}}^{\otimes}(\omega) = \varprojlim_X \underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X}).$$

The second equality holds because every finite collection of representations and morphisms lies in some $\mathcal{C}_X$.

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